

Derbyshire Shared Care Pathology Guidelines

Overnight Dexamethasone Test for Primary Care

This screening test for Cushing's syndrome is suitable for out-patients or in-patients, but is best done on an out-patient basis. Cushing's syndrome is relatively rare and false positive results can occur in a variety of situations (see INTERPRETATION). Abnormal results with this test should be followed by further investigation by an endocrinologist.

Dexamethasone is a synthetic steroid with 25 times the glucocorticoid activity of cortisol. It does not interfere with cortisol measurement. Dexamethasone leads to suppression of adrenal cortisol secretion in normal individuals. In patients with Cushing's syndrome there is incomplete suppression.

This test is performed as an overnight suppression test, using a single dexamethasone dose of 1 mg.

INDICATIONS

As a first-line screening test to exclude Cushing's syndrome

CONTRAINDICATIONS

None

SIDE EFFECTS

None

PRECAUTIONS

This procedure may be contraindicated in patients with active peptic ulcer disease.

PREPARATION

Planning

Out-patients should be fully briefed on when to take the dexamethasone and to return the following morning for a blood test at 9am. Patients should be instructed not to take the dexamethasone on a Friday or Saturday night.

Equipment

2 x 500 microgram tablets dexamethasone

PROCEDURE

Out-Patients

Supply the patient with a 1 mg dose of dexamethasone (2 x 500 microgram tablets). Instruct the patient to take both tablets together, between 23:00 and midnight on the evening before their blood test, which should be planned for the following morning between 09:00 and 09:30.

The morning after the dexamethasone is taken, a single sample for serum cortisol should be collected into a SST, yellow top vacutainer tube. Send the blood sample with completed request form, for "post-dexamethasone cortisol", to the Chemical Pathology Department.

CHISCP45

Revision:4

Date of issue: Feb 2026

Expiry date:28th Feb 2029

Approved by:Julia Forsyth

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INTERPRETATION

Normal suppression is indicated by a 09:00 serum cortisol concentration of less than 50 nmol/L.

Note that failure of cortisol suppression can also occur in:

1. Severe endogenous depression - these patients may have abnormal circadian rhythm.
2. Alcoholism (pseudo-Cushing's syndrome)
3. Severe stressful illness/infection - the test should not be performed in this situation.
4. Hepatic enzyme-inducing drugs - (phenytoin, phenobarbitone, rifampicin, etc). These may cause more rapid metabolism of dexamethasone, such that normal suppression of the pituitary-adrenal axis fails.
5. Pregnancy or Oestrogen therapy - induces high levels of cortisol binding globulin, such that the test is not interpretable.
6. Failure to take dexamethasone correctly - check with patient.
7. Glucocorticoid resistance syndrome - rare, familial disorder and patients are not Cushingoid.
8. Obesity
9. Renal failure

NOTES

Normal suppression makes a diagnosis of Cushing's Syndrome unlikely. The specificity of the test is 87%.

Failure of suppression does not confirm Cushing's Syndrome. Abnormal results with this test should be followed by further investigation under the supervision of an endocrinologist.

If there is still a high clinical index of suspicion, further tests may be necessary to exclude conditions such as cyclical Cushing's. Discuss with an endocrinologist.

Please give full drug history when asking for cortisol measurements.

Document controls

Document title	Overnight Dexamethasone Test for Primary Care		
Revision	4	Document Reference	CHISCP36
Reason for review	Routine review		
Review history	Version/Date	Authors	Reason for review
	V1 Nov 2015	Dr R Stanworth, Mrs H Seddon	New Document
	V2 Jan 2019	Biochemistry/Endocrinology MDT meeting	Routine review
	V3 Jan 2022	Biochemistry/Endocrinology MDT meeting	Routine review
Approval	Date	Name(s)	Designation
Secondary Care UHDB	12.12.25	Biochemistry/Endocrinology MDT group	Consultant Endocrinologists
Secondary Care CRH	03.02.26	Dr A Pillai, Dr R Robinson, Dr P Shillo Biochemistry/Endocrinology MDT group	Consultant Endocrinologists
Primary Care	03.02.26	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	03.02.26	Ms S Knowles	Consultant Clinical Scientist